



Mohamed Oubih

Mechanical Engineer

Based in Bratislava, Slovakia · Italian (EU) citizen · Open to relocation

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4 YEARS

HARDWARE DEVELOPMENT

AMS-02

ISS DETECTOR PROGRAMME

20+

LAUNCH CAMPAIGNS

50+

PRODUCTION DRAWINGS

SUMMARY

Mechanical engineer who takes hardware from requirements through CAD, prototyping, supplier release, and environmental testing. Designed alignment and bonding tooling for the AMS-02 Layer 0 silicon-tracker upgrade, produced 50+ GD&T production drawings in automotive, and is currently the sole mechanical engineer on an ultralight radiosonde and superpressure balloon system. Strongest in lightweight sealing, low-temperature mechanisms, and developing test systems to validate them.

EXPERIENCE

Mechanical & Aerospace Engineer

August 2025 – Present · 1 yr 1 mo

Noove s.r.o. · Bratislava, Slovakia

- **Sole mechanical engineer on Project Picoballoon**, owning all mechanical hardware for the FD-11 radiosonde (~20 g) and the superpressure balloon system (26 µm TritaX envelope): design, drawings, GD&T, BOMs, supplier coordination and release, assembly, and field-test support across 20+ launch campaigns.
- **Developed and characterized a miniature passive pressure-relief valve** through 3 iterations, replacing a direct-acting concept with a **diaphragm architecture** that sets the pressure-sensing area independently of the seat bore, raising design seat contact stress from ~0.04 MPa to ~5 MPa (>100×) at the same cracking pressure; achieved a tunable **1.4–6.5 kPa** range, linear preload response across six configurations ($R^2 = 0.9996$), and derived the flight **venting requirement** from mission telemetry (helium mass flow at 220 K, 17 kPa absolute).
- **Identified two independent low-temperature failure modes**: Silicone **O-ring static seal** leakage at $-30 \pm 3 \text{ }^\circ\text{C}$, and a **33% rise in cracking pressure** traced to reduced effective diaphragm area rather than additive stiction (cold/ambient ratio constant at 1.327 ± 0.002 across a 2.6× preload range) — redirecting the redesign from seat to diaphragm.
- Built the **Python/PyQt5 fluidic test platform** used for all valve characterisation: closed-loop differential-pressure control against ambient drift through a networked pressure calibrator, Bronkhorst mass-flow measurement, automated staircase and endurance profiles, and logging and analysis of datasets containing 20k+ samples.
- Owned **FD-11 radiosonde enclosure** development across 7 CATIA/SLA iterations to manufacturing release with outsourced SLA (JLC3DP), covering drawings, GD&T, BOMs, defect-rate analysis and corrective redesign, mass budget, PCB/battery integration, and assembly repeatability.
- Designed **assembly and calibration hardware** for flexible humidity-sensor boards, including SLA fixtures and 0.2 mm laser-cut positioning masks, to improve sensor placement repeatability and reduce operator dependence.

Mechanical Design Engineer

May 2023 – July 2025 · 2 yrs 3 mos

OPmobility · Bratislava, Slovakia

- **Modelled injection-moulded automotive exterior assemblies in CATIA V5** (bumper systems, trim interfaces, brackets, attachment features), applying DFM rules for draft, wall thickness, ribs, snap-fits, tooling direction, parting strategy, undercuts, and sink-mark risk.
- Created and maintained **50+ production drawings** with GD&T, datum schemes, tolerance analysis, and material/process notes to ANSI/ASME standards; managed CAD maturity across multiple vehicle programmes through clearance checks, interface reviews, and assembly-feasibility studies.
- Supported prototype, dimensional, and validation activities for bumper systems on Škoda, Opel, and Citroën platforms, implementing CAD corrections from tooling, quality, manufacturing, and test feedback.

Research Fellow

August 2022 – May 2023 · 10 mos

University of Perugia · Terni, Italy

- Designed **mechanical integration tooling for the AMS-02 Layer 0 silicon tracker upgrade**, an ISS-based particle-physics detector programme, focused on alignment and adhesive bonding of fragile silicon-sensor subassemblies onto carbon-composite support structures.
- Developed precision alignment fixtures using pin-hole/slot datum interfaces and support buttons, controlling subassembly position within approximately **100 µm across a 1100 × 1200 mm** integration fixture.
- Performed tolerance-stack and datum analysis across tooling and support interfaces, reducing worst-case alignment error from **~120 µm to ~93 µm** through tighter fits and machining specification.
- **Optimised the composite support strategy using ANSYS FEM/FEA**, increasing the support-button layout from 9 to 13 points and reducing predicted self-weight deformation from **~62 µm to ~26 µm** within the deformation budget.
- Built and validated a reduced proof-of-concept prototype, and conducted **PEEK / 3M 2216 epoxy shear testing** including surface preparation, controlled curing, bond-line measurement, and failure-mode analysis.

SKILLS

CAD / Mechanical Design — CATIA V5, SolidWorks, Onshape, solid and surface design, complex assemblies, Geometric Dimensioning and Tolerancing, tolerance analysis, 2D/3D production drawings

Testing / Validation — Pressure, leak, and burst testing; low-temperature and cold-chamber testing to –30 °C; vibration (random, sine, shock); thermal-vacuum chamber; strain gauges; thermocouples; mass-flow and leak-rate measurement

Analysis / Simulation — ANSYS FEM/FEA (stress, deformation, material selection), FMEA, DFM (injection molding, SLA)

Prototyping / Manufacturing — SLA (Formlabs Form 4), FDM, outsourced SLA (JLC3DP), CNC and machined-part sourcing, laser-cut fixtures, assembly tooling, manufacturing-release documentation

Automation / Data — Python, PyQt5, MATLAB, pump / flowmeter / pressure-sensor / valve integration, automated test sequencing and data analysis

Materials / Processes — PTFE, PEEK, Kapton/polyimide, elastomers, SLA resins, carbon-fibre honeycomb, aluminium, adhesive bonding

LANGUAGES

Italian (native) · English (fluent) · Arabic (intermediate) · Slovak (beginner)

EDUCATION

MSc Mechanical Engineering

2020 – 2022 · 2 yrs

University of Perugia · *Cum Laude*

Thesis: design, characterisation, and testing of the alignment and gluing system for integration of silicon cosmic-ray sensors into the AMS-02 Layer 0 tracker upgrade.

BSc Mechanical Engineering

2016 – 2020 · 4 yrs

University of Perugia

Thesis: Measurement of the magnetic characteristics of Manganese-Zinc ferrite cores